

FIE401 - Assignment 2

Estimating market risk premia

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Overview

A central question in asset pricing is how to value an asset. Put differently, how to quantify the risk-return trade-off. Development of the CAPM allowed economists to formally quantify the risk (market risk = β) and the reward for carrying it (market risk premia = λ).

In this assignment, you are going to estimate market risk (i.e., β) of 25 portfolios sorted on size and book-to-market ratio based on the CAPM model. Afterwards, you are going to estimate market risk premia implied by the CAPM model. This approach is known as Fama-MacBeth (1973) approach.

Formalities

This assignment will be handed out on 10.03.2026 at 12:00 and has to be submitted no later than the **17.03.2026 at 09:00**. Submit your commented coding file (file extension: ".R") and a pdf file (file extension: ".pdf") including the numerical results and answers to questions posted.

Please comment your code shortly so that a reader can reconstruct your thinking. You do not need to explain the used functions. You do not need to describe your coding in the pdf file. Please keep your answers very brief. **Code without comments will automatically result in a "Fail" grade.**

On Canvas, you will find a guideline of how to make good tables. Note that in the last lecture of this course, we will cover the subject of how to present research in detail.

After submission, the assignments will be randomly redistributed. Read your peer's work carefully and comment within Canvas no later than **24.03.2026 at 09:00**. Your comments are meant to improve the work of your peers, so use constructive feedback. See peer-review guidelines for details. **Insufficient quality of peer-review or no peer-review will result in a "Fail" grade, regardless of the quality of your own analysis.**

In order to make it easy for us to allocate individual assignment submission as well as peer review to groups, please follow following file name structure. **For the assignment submission: "Assignment2_group_XX.R/pdf"**. "XX" indicates the group number that submitted the assignment. **Please submit the peer review as a comment on CANVAS.**

Please do not include any personal information such as name, student number, etc. in the submitted files.

Note that in order to complete the assignment, you need to submit your own work as well as complete the peer review. Both aspects will be reviewed.

Work together in groups of 3-4 people.

Submit your assignment even if you do not finish all tasks.

Note that some concepts are introduced for the first time in this assignment. Please use online resources actively.

We touched on almost all functions you have to use to solve this assignment. If you are unsure how to use a function, either use R's own documentation (type `?command`) or use www.stackoverflow.com. We encourage you to code this assignment yourselves, and not to use purpose-made solutions or packages as provided on the internet.

Before starting to code, try to separate the task into smaller pieces.

Data description

File `25_portfolios_on_size_BM.Rdata` contains monthly returns on 25 portfolios sorted on size (ME) and book-to-market ratio (BM) as well as market excess return (`MktRF`) and risk free rate (`RF`) from January 1980 till December 2017. ME1 contains the smallest stocks, while ME5 contains the largest stocks. BM1 contains stocks with the lowest book-to-market ratio, while BM5 contains stocks with the highest book-to-market ratio. Data come from Kenneth French's website.

Tasks

First look at the data

- Load the data and investigate it.

Fama-MacBeth approach: Part 1

- For each portfolio p estimate market risk (β_p) implied by CAPM:

$$R_{pt} - RF_t = \alpha_p + \beta_p \times MktRF_t + u_{pt}$$

- This step should result in having 25 market betas ($\hat{\beta}_p$) for each of the 25 portfolios from 25 individual regressions. Note, that I provide a coding hint on how to use loops and save regression coefficients at the end of the assignment.
- Plot the market beta of each individual portfolio on the x-axis against the average return over the period on the y-axis. Does market risk explain differences in return? Does the sign of the relationship (positive/negative) correspond to your expectations?

Fama-MacBeth approach: Part 2

- Estimate market risk premia (λ).
- For each month run cross-sectional regression of portfolio returns on $\hat{\beta}_p$:

$$R_{pt} = a_t + \lambda_t \times \hat{\beta}_p + \omega_{pt}$$

- This step should result in having 456 risk premias ($\hat{\lambda}_t$) for each of the 456 months in the sample from 456 individual regressions. Note, that I provide a coding hint on how to use loops and save regression coefficients at the end of the assignment.
- Plot time-series of monthly $\hat{\lambda}_t$.

Fama-MacBeth approach: Part 3

- Compute average of monthly $\hat{\lambda}_t$ and test if it is significantly different from 0.
- What do you conclude regarding the price of market risk?
- Which bias might you be concerned about in this analysis, and how would it affect your results?

Coding hint

In two tasks of this assignment, you are asked to run sequential regressions and save the coefficients. Here I provide a general **pseudo-code** of how this can be done.

```
# object to save 25 coefficients
coef.vec <- rep(NA, 25)

# iteration
for(i in ...){
  # construct relevant data
  rel.data <- ...

  # regression
  fit <- lm(..., data = rel.data)

  # save coefficient
  coef.vec[i] <- fit$coef[...]
}
```

Note, feel free to use more elegant and faster coding approaches.

References

Fama, E. F., MacBeth, J. D., 1973, Risk, Return and Equilibrium: Empirical Tests. Journal of Political Economy 81 (3), pp. 607-636

```
#####
#                                     #
# Suggested code solution for Assignment 2 #
#                                     #
#####

# load necessary packages
require(dplyr)
require(tibble)
require(ggplot2)

# set working directory
setwd("C:/Users/s13163/Dropbox/FIE401/data/data_assignments/")

# load data on 25 size and BM portfolios
load("25_portfolios_on_size_and_BM.RData")
summary(DATA)
```

##	Date	ME1.BM1	ME1.BM2	ME1.BM3
##	Min. :198001	Min. :-34.2172	Min. :-30.948	Min. :-28.765
##	1st Qu.:198907	1st Qu.: -4.0584	1st Qu.: -2.452	1st Qu.: -1.970
##	Median :199857	Median : 1.0702	Median : 1.709	Median : 1.443
##	Mean :199857	Mean : 0.4692	Mean : 1.163	Mean : 1.204
##	3rd Qu.:200806	3rd Qu.: 4.8624	3rd Qu.: 5.082	3rd Qu.: 4.581
##	Max. :201712	Max. : 38.9062	Max. : 41.052	Max. : 21.863
##	ME1.BM4	ME1.BM5	ME2.BM1	ME2.BM2
##	Min. :-28.890	Min. :-28.869	Min. :-32.7265	Min. :-31.658
##	1st Qu.: -1.366	1st Qu.: -1.230	1st Qu.: -3.7388	1st Qu.: -2.097
##	Median : 1.493	Median : 1.569	Median : 1.5458	Median : 1.575
##	Mean : 1.391	Mean : 1.430	Mean : 0.8792	Mean : 1.218
##	3rd Qu.: 4.310	3rd Qu.: 4.718	3rd Qu.: 5.2141	3rd Qu.: 4.704
##	Max. : 25.400	Max. : 17.343	Max. : 28.1757	Max. : 17.816
##	ME2.BM3	ME2.BM4	ME2.BM5	ME3.BM1
##	Min. :-28.402	Min. :-25.036	Min. :-28.837	Min. :-29.7258
##	1st Qu.: -1.303	1st Qu.: -1.459	1st Qu.: -1.522	1st Qu.: -3.0984
##	Median : 1.631	Median : 1.706	Median : 1.938	Median : 1.7686
##	Mean : 1.287	Mean : 1.324	Mean : 1.289	Mean : 0.9763
##	3rd Qu.: 4.294	3rd Qu.: 4.604	3rd Qu.: 4.694	3rd Qu.: 5.2222
##	Max. : 16.346	Max. : 16.254	Max. : 19.150	Max. : 24.6065
##	ME3.BM2	ME3.BM3	ME3.BM4	ME3.BM5
##	Min. :-29.052	Min. :-24.338	Min. :-23.031	Min. :-26.119
##	1st Qu.: -1.802	1st Qu.: -1.491	1st Qu.: -1.174	1st Qu.: -1.090
##	Median : 1.543	Median : 1.562	Median : 1.483	Median : 1.705
##	Mean : 1.249	Mean : 1.188	Mean : 1.268	Mean : 1.438
##	3rd Qu.: 4.381	3rd Qu.: 4.159	3rd Qu.: 4.071	3rd Qu.: 4.342
##	Max. : 18.189	Max. : 17.161	Max. : 16.245	Max. : 17.392
##	ME4.BM1	ME4.BM2	ME4.BM3	ME4.BM4
##	Min. :-25.940	Min. :-28.829	Min. :-25.155	Min. :-22.403
##	1st Qu.: -2.275	1st Qu.: -1.674	1st Qu.: -1.306	1st Qu.: -1.418
##	Median : 1.442	Median : 1.407	Median : 1.530	Median : 1.504
##	Mean : 1.162	Mean : 1.164	Mean : 1.136	Mean : 1.229
##	3rd Qu.: 4.704	3rd Qu.: 4.178	3rd Qu.: 4.002	3rd Qu.: 3.930
##	Max. : 26.222	Max. : 15.856	Max. : 16.881	Max. : 14.378
##	ME4.BM5	ME5.BM1	ME5.BM2	ME5.BM3

```
## Min.      :-23.840   Min.      :-21.639   Min.      :-22.420   Min.      :-21.709
## 1st Qu.: -1.524   1st Qu.: -1.449   1st Qu.: -1.490   1st Qu.: -1.460
## Median :  1.614   Median :  1.167   Median :  1.428   Median :  1.417
## Mean    :  1.237   Mean    :  1.048   Mean    :  1.081   Mean    :  1.072
## 3rd Qu.:  4.480   3rd Qu.:  3.927   3rd Qu.:  3.709   3rd Qu.:  3.776
## Max.    : 17.032   Max.    : 15.356   Max.    : 16.531   Max.    : 19.234
## ME5.BM4      ME5.BM5      MktRF      RF
## Min.      :-27.0162   Min.      :-18.992   Min.      :-23.2400   Min.      :0.0000
## 1st Qu.: -1.4078   1st Qu.: -1.458   1st Qu.: -1.9250   1st Qu.:0.0800
## Median :  1.3216   Median :  1.549   Median :  1.0900   Median :0.3750
## Mean    :  0.9066   Mean    :  1.193   Mean    :  0.6716   Mean    :0.3555
## 3rd Qu.:  3.5417   3rd Qu.:  4.346   3rd Qu.:  3.4900   3rd Qu.:0.5100
## Max.    : 16.0942   Max.    : 23.626   Max.    : 12.4700   Max.    :1.3500
```

```
# FMB Part 1
```

```
# object to save 25 coefficients
```

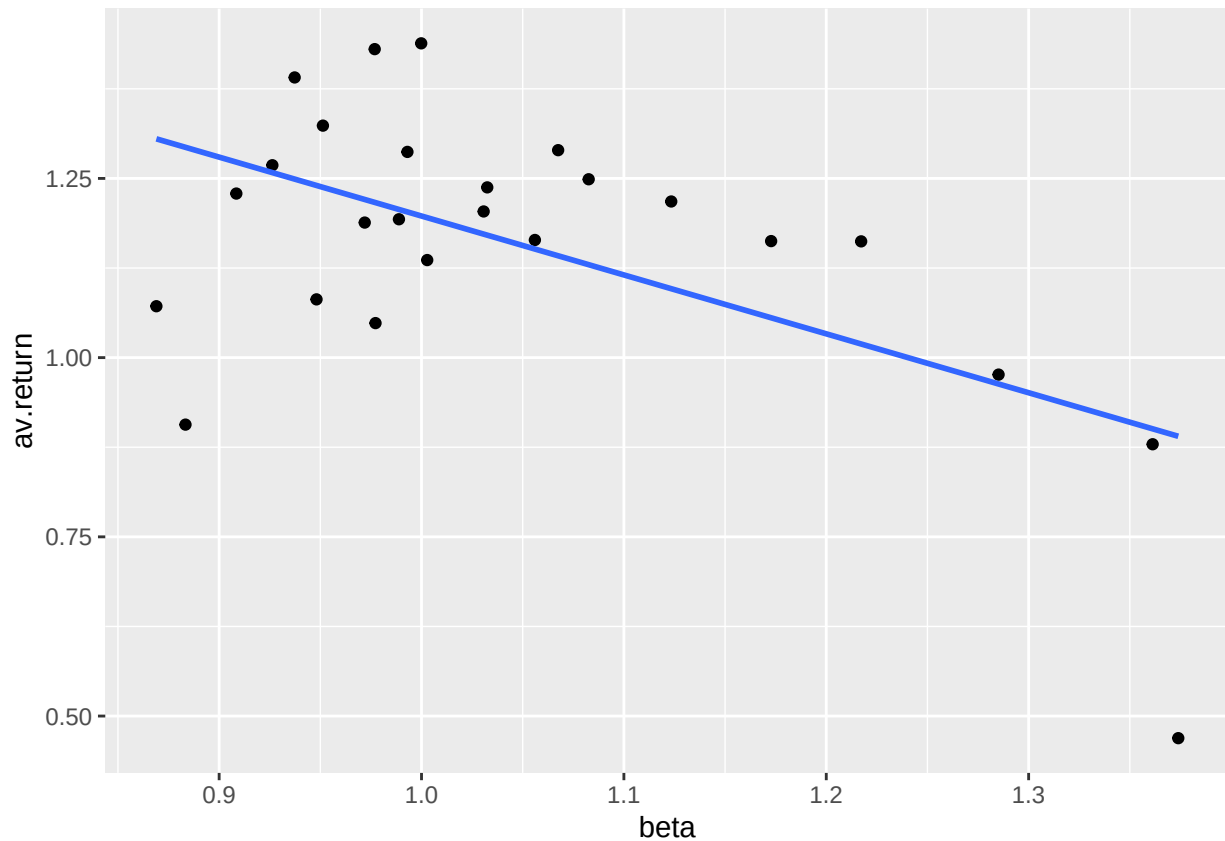
```
beta <- rep(NA, 25)
```

```
for (i in 2:26)
```

```
{
  #extract columns i (returns for portfolio[i-1]),
  #27 (MktRF) and 28 (RF) from the original data-frame
  DATAi<-DATA[,c(i,27,28)];
  #generate excess returns for portfolio [i-1]
  DATAi$ExRet<-DATAi[,1]-DATAi[,3];
  #estimate CAPM
  fit<-lm(ExRet~MktRF,data=DATAi)
  #store the results in the data-frame RESULT
  beta[(i-1)]<- fit$coef[2]
  #remove DATAi and fit
  rm(DATAi,fit);
}
```

```
# plot: x-axis beta, y-axis average return
```

```
ggplot(data = tibble(beta = beta,
                      av.return = colMeans(DATA[,2:26])),
       aes(x = beta, y = av.return)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y~x, se =F)
```

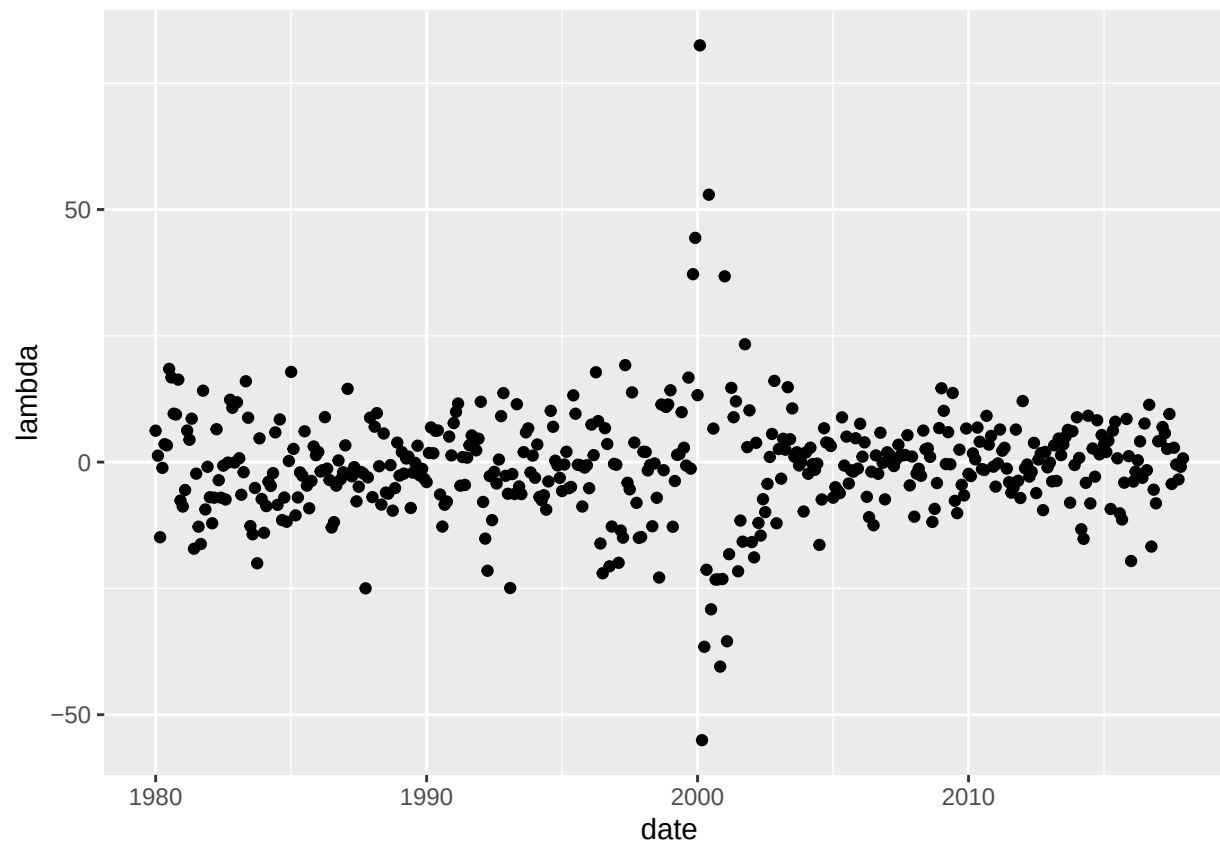


```
# FMB Part 2

# object to save 456 coefficients
lambda <- rep(NA, 456)

for (i in 1:456)
{
  #extract row i and columns from 2 till 26 from the original data frame
  #i.e., returns in month i for 25 portfolios
  #and combine it with beta estimates
  DATAi<- tibble(ret=t(DATA[i,c(2:26)]),beta=beta);
  #estimate the regression
  fit<-lm(ret~beta,data=DATAi)
  #store the results in the data-frame RESULT
  lambda[i]<- fit$coef[2]
  #remove DATAi and fit
  rm(DATAi,fit);
}

# plot: x-axis date, y-axis risk premia
ggplot(data = tibble(lambda =lambda,
                      date = seq(from=as.Date("1980-01-01",format="%Y-%m-%d"),
                                length.out=456, by='1 mon')),
  aes(x = date, y = lambda)) +
  geom_point()
```



```
# FMB Part 3
#Note that in theory risk premia should be positive
#since taking more risk should be compensated with higher return.
#Hence, I perform a one-sided t-test

t.test(lambda,alternative="greater")

##
## One Sample t-test
##
## data:  lambda
## t = -1.6245, df = 455, p-value = 0.9475
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
##  -1.655453      Inf
## sample estimates:
## mean of x
## -0.821731

# One might worry about measurement error in the independent variable, since the market beta
# needs to be estimated
```